

GBGHMVNNEXSONJEN - COMPLETE SINGLE PDF GUIDE

From Zero to BUILD SUCCESS - Windows 11 | Java + Maven + Nexus + SonarQube + Jenkins + GitHub
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PART 1: SIMPLE STORY FOR 6-YEAR-OLD - What is this project?

Think of Toy Factory:

- You draw toy car (code in App.java)
- GitHub = Your drawing book (keeps drawings safe)
- Jenkins = Boss Robot (shouts BUILD when new drawing comes)
- Maven = Builder Machine (makes real toy JAR)
- Nexus = Big Shelf (stores toys) at <http://localhost:8081>
- SonarQube = Doctor (checks toy is safe) at <http://localhost:9000>

Flow: You draw -> GitHub -> Jenkins sees -> Maven builds -> Nexus stores -> SonarQube checks -> SUCCESS!

GBGHMVNNEXSONJEN letters: G=GitHub, B+M=Maven Build, N=Nexus, SON=SonarQube, JEN=Jenkins

PART 2: HOW WE INSTALLED EVERYTHING ON WINDOWS 11

A) JAVA 21 - Electricity for all machines

```
Download: https://adoptium.net -> JDK 21 Windows x64 MSI Install Location: C:\devops\java\jdk-21 Environment  
Variables: JAVA_HOME = C:\devops\java\jdk-21 Path -> Add %JAVA_HOME%\bin Verify in CMD / Git Bash: java -version ->  
openjdk 21 javac -version -> javac 21
```

B) MAVEN 3.9.14 - Builder Machine

```
Download: https://maven.apache.org/download.cgi -> Binary zip Extract to: E:\maven\apache-maven-3.9.14 Environment  
Variables: M2_HOME = E:\maven\apache-maven-3.9.14 MAVEN_HOME = E:\maven\apache-maven-3.9.14 Path -> Add %M2_HOME%\bin  
Verify: mvn -version -> Apache Maven 3.9.14 + Java 21 Create secret file for Nexus: Location:  
C:\Users\AFZALBABA\.m2\settings.xml Content: <settings> <servers>  
<server><id>maven-snapshots</id><username>admin</username><password>Admin123</password></server>  
<server><id>maven-releases</id><username>admin</username><password>Admin123</password></server> </servers> </settings>
```

C) NEXUS 3.x - Big Shelf for Toys

```
Download: https://www.sonatype.com/products/nexus-repository -> OSS Extract: C:\devops\nexus\nexus-3.75 Install &  
Start (Admin CMD): cd C:\devops\nexus\nexus-3.75\bin nexus.exe /install nexus nexus.exe /start nexus Wait 2 min -> Open  
http://localhost:8081 First password: C:\devops\nexus\sonatype-work\nexus3\admin.password Login admin / password ->  
Create hosted repos: maven-snapshots (Version policy: Snapshot) maven-releases (Version policy: Release)
```

D) SONARQUBE 10.x - Doctor Checker

```
Download: https://www.sonarqube.org/downloads/ -> Community Edition Extract: C:\devops\sonarqube-10.6 Start (CMD): cd  
C:\devops\sonarqube-10.6\bin\windows-x86-64 StartSonar.bat Open: http://localhost:9000 -> admin / admin Generate  
Token: Top Right -> My Account -> Security -> Tokens Name: jenkins, Type: Global Analysis Token -> Generate Copy:  
sqp_XXXXXXXXXXXXXXXXXXXXX SAVE IT!
```

E) JENKINS - Boss Robot

```
Download: https://www.jenkins.io/download/ -> Windows MSI Install -> http://localhost:8080 Unlock:  
C:\ProgramData\Jenkins\.jenkins\secrets\initialAdminPassword Install Plugins: Suggested + Git + Maven Integration  
CRITICAL FIX (This was our ERROR 3): Manage Jenkins -> Tools -> Git installations -> Add Git Name: Default Path:  
C:\Program Files\Git\bin\git.exe If not exist: C:\Program Files\Git\cmd\git.exe SAVE -> Restart Jenkins: net stop  
jenkins net start jenkins Add JDK & Maven tools: Manage Jenkins -> Tools -> JDK installations -> Add JDK Name: jdk21,  
JAVA_HOME: C:\devops\java\jdk-21 Maven installations -> Add Maven Name: maven3, MAVEN_HOME:  
E:\maven\apache-maven-3.9.14
```

PART 3: ALL FILE NAMES & LOCATIONS - TREASURE MAP

File	Location	What it is
Project Root	E:\project-GBGHMVNNEXSONJEN	Factory
App.java	src\main\java\com\gbghmvnnexsonjen\App.java	Toy Drawing
pom.xml	E:\project-GBGHMVNNEXSONJEN\pom.xml	Recipe Book
Jenkinsfile	E:\project-GBGHMVNNEXSONJEN\Jenkinsfile	Boss To-Do
README.md	E:\project-GBGHMVNNEXSONJEN\README.md	Notice Board
settings.xml	C:\Users\AFZALBABA\m2\settings.xml	Secret Key
Maven	E:\maven\apache-maven-3.9.14	Builder
Java	C:\devops\java\jdk-21	Electricity
Jenkins Home	C:\ProgramData\Jenkins\jenkins	Boss Brain
Jenkins Workspace	C:\ProgramData\Jenkins\jenkins\workspace\GBGHMVNNEXSONJEN	SON JEN
Built JAR	E:\project-GBGHMVNNEXSONJEN\target*.jar	Final Toy

PART 4: GIT BASH COMMANDS - How We Created Local Repo & Artifact (PuTTY Style)

Copy-paste in Git Bash (Right click in E:\ -> Git Bash Here):

```
$ mkdir project-GBGHMVNNEXSONJEN $ cd project-GBGHMVNNEXSONJEN $ mkdir -p src/main/java/com/gbghmvnnexsonjen $ cat >
src/main/java/com/gbghmvnnexsonjen/App.java << 'EOF' package com.gbghmvnnexsonjen; public class App { public static
void main(String[] args){ System.out.println("Hello DevOps - GBGHMVNNEXSONJEN!"); } } EOF $ cat > pom.xml << 'EOF'
<project xmlns="http://maven.apache.org/POM/4.0.0"> <modelVersion>4.0.0</modelVersion>
<groupId>com.gbghmvnnexsonjen</groupId> <artifactId>project-GBGHMVNNEXSONJEN</artifactId>
<version>1.0-SNAPSHOT</version> <properties> <maven.compiler.source>21</maven.compiler.source>
<maven.compiler.target>21</maven.compiler.target> </properties> <distributionManagement> <snapshotRepository>
<id>maven-snapshots</id> <url>http://localhost:8081/repository/maven-snapshots</url>
</snapshotRepository> <repository> <id>maven-releases</id> <url>http://localhost:8081/repository/maven-releases</url>
</repository> </distributionManagement> </project> EOF $ cat > Jenkinsfile << 'EOF' pipeline { agent any tools { jdk
'jdk21'; maven 'maven3' } stages { stage('Build') { steps { bat 'cd /d E:\project-GBGHMVNNEXSONJEN && mvn clean package
-DskipTests' } } stage('Deploy to Nexus') { steps { bat 'cd /d E:\project-GBGHMVNNEXSONJEN && mvn deploy -DskipTests' }
} stage('SonarQube Analysis') { steps { bat 'cd /d E:\project-GBGHMVNNEXSONJEN && mvn
org.sonarsource.scanner.maven:sonar-maven-plugin:5.0.0.4389:sonar -Dsonar.projectKey=project-GBGHMVNNEXSONJEN
-Dsonar.host.url=http://localhost:9000 -Dsonar.token=sqp_YOUR_REAL_TOKEN -Dsonar.scanner.skipJreProvisioning=true' } }
} } EOF $ mvn clean package [INFO] BUILD SUCCESS -> target/project-GBGHMVNNEXSONJEN-1.0-SNAPSHOT.jar $ mvn deploy
-DskipTests [INFO] BUILD SUCCESS -> Uploaded to Nexus! $ git init $ git add . $ git commit -m "Initial commit" $ git
remote add origin https://github.com/afzal-baba/project-GBGHMVNNEXSONJEN.git $ git branch -M main $ git push -u origin
main # If GitHub ahead (you edited on web): $ git pull origin main --rebase $ git push origin main # If nested folder
bug: $ rm -rf project-GBGHMVNNEXSONJEN $ git rm --cached project-GBGHMVNNEXSONJEN $ git add . && git commit -m "fix" &&
git push origin main --force
```

PART 5: ALL ERRORS WE FACED & HOW WE FIXED (Detective Story)

ERROR 1: pom.xml not found

Cause (Kid): You ran mvn in E:\ not inside project

Fix: cd /e/project-GBGHMVNNEXSONJEN then mvn clean package - Go inside kitchen before cooking!

ERROR 2: Failed to connect to repository / git command

Cause (Kid): Jenkins no GitHub password

Fix: GitHub -> Settings -> Developer Settings -> Personal Access Tokens -> Tokens classic -> Generate -> repo scope -> Copy ghp_xxx -> Jenkins Add Credentials -> Username with password -> Username afzal-baba, Password ghp_xxx, ID github-token

ERROR 3: CreateProcess error=87 Cannot run program ""

Cause (Kid): Jenkins doesn't know where git.exe is

Fix: Manage Jenkins -> Tools -> Git installations -> Add Git -> Name Default Path C:\Program Files\Git\bin\git.exe -> Save -> net stop jenkins & net start jenkins

ERROR 4: 401 Unauthorized SonarQube / sqp_YOUR_TOKEN

Cause (Kid): Fake token or SonarCloud token

Fix: Go to http://localhost:9000 -> My Account -> Security -> Generate Token -> Global Analysis Token -> Copy sqp_... -> Paste in Jenkinsfile, NOT sonarcloud.io token

ERROR 5: sonar.organization required

Cause (Kid): Used SonarCloud token on local SonarQube

Fix: Use ONLY token from localhost:9000, keep sonar.host.url=http://localhost:9000, don't use sonarcloud.io

ERROR 6: Branch master not found

Cause (Kid): GitHub now uses main

Fix: Jenkins job -> Configure -> Branches to build -> Branch Specifier: */main not */master

ERROR 7: Updates rejected fetch first

Cause (Kid): Edited file on GitHub web, laptop old

Fix: git pull origin main --rebase then git push origin main OR git push --force

ERROR 8: mode 160000 nested repo

Cause (Kid): Box inside box - cloned inside itself

Fix: rm -rf nested folder, git rm --cached folder, commit, push --force

PART 6: JENKINS PIPELINE EXPLAINED

```
pipeline { agent any tools { jdk 'jdk21'; maven 'maven3' } stages { stage('Build') { // Makes JAR file bat 'cd /d E:\project-GBGHMVNNEXSONJEN && mvn clean package -DskipTests' } stage('Deploy to Nexus') { // Puts JAR on shelf http://localhost:8081 bat 'cd /d E:\project-GBGHMVNNEXSONJEN && mvn deploy -DskipTests' } stage('SonarQube Analysis') { // Doctor checks at http://localhost:9000 bat 'cd /d E:\project-GBGHMVNNEXSONJEN && mvn sonar:sonar -Dsonar.projectKey=project-GBGHMVNNEXSONJEN -Dsonar.host.url=http://localhost:9000 -Dsonar.token=sqp_REAL -Dsonar.scanner.skipJreProvisioning=true' } } } # Result: BUILD SUCCESS - All 3 GREEN!
```

PART 7: COLORFUL PROFESSIONAL README.md FOR GITHUB (Copy-Paste)

Create README.md in E:\project-GBGHMVNNEXSONJEN and paste this:

```
# ■ GBGHMVNNEXSONJEN - Enterprise DevOps CI/CD [![Jenkins SUCCESS](https://img.shields.io/badge/Jenkins-BUILD%20SUCCESS-brightgreen?style=for-the-badge)](https://github.com/afzal-baba/project-GBGHMVNNEXSONJEN) [![Maven](https://img.shields.io/badge/Maven-3.9.14-C71A36?style=for-the-badge&logo=apachemaven)](https://maven.apache.org/) [![Java](https://img.shields.io/badge/Java-21-ED8B00?style=for-the-badge&logo=openjdk)](https://openjdk.org/) [![Nexus](https://img.shields.io/badge/Nexus-8081-0FAAFF?style=for-the-badge&logo=sonatype)](http://localhost:8081) [![SonarQube](https://img.shields.io/badge/SonarQube-9000-4E9BCD?style=for-the-badge&logo=sonarqube)](http://localhost:9000) > Full Lifecycle on Windows 11: GitHub | Maven | Nexus | SonarQube | Jenkins ## ■ Kid Story G=GitHub (Notebook) | M=Maven (Builder) | N=Nexus (Shelf) | SON=SonarQube (Doctor) | JEN=Jenkins (Boss Robot)

Flow: Dev -> GitHub -> Jenkins -> Maven -> Nexus -> SonarQube -> SUCCESS ## ■■ Install Locations (Windows 11) - Java 21: C:\devops\java\jdk-21 - Maven 3.9.14: E:\maven\apache-maven-3.9.14 - Maven Settings: C:\Users\%USERNAME%\m2\settings.xml - Nexus: C:\devops\nexus + http://localhost:8081 - SonarQube: C:\devops\sonarqube + http://localhost:9000 - Jenkins: C:\ProgramData\Jenkins\jenkins + http://localhost:8080 - Project: E:\project-GBGHMVNNEXSONJEN ## ■ Quick Start (Git Bash) git clone https://github.com/afzal-baba/project-GBGHMVNNEXSONJEN.git cd project-GBGHMVNNEXSONJEN mvn clean package mvn deploy # Check Nexus and SonarQube! ## ■ Artifact com.gbghmvnnexsonjen:project-GBGHMVNNEXSONJEN:1.0-SNAPSHOT -> Nexus snapshots ## ■■■ Author afzal-baba | github.com/afzal-baba ## ■ License MIT - Learning Project
```

PART 8: FINAL CHECK & INTERVIEW ANSWER

Verify After BUILD SUCCESS:

1. Nexus: http://localhost:8081 -> Browse -> maven-snapshots -> com -> gbghmvnnexsonjen -> project-GBGHMVNNEXSONJEN -> 1.0-SNAPSHOT -> .jar exists!
2. SonarQube: http://localhost:9000 -> Projects -> project-GBGHMVNNEXSONJEN -> Issues, Coverage, Quality Gate PASSED
3. Jenkins: http://localhost:8080/job/GBGHMVNNEXSONJEN/ -> Last build GREEN
4. Local: E:\project-GBGHMVNNEXSONJEN\target\project-GBGHMVNNEXSONJEN-1.0-SNAPSHOT.jar

30 Sec Interview Story:

"I built end-to-end CI/CD on Windows 11 from scratch. Installed Java 21, Maven 3.9, Nexus, SonarQube, Jenkins. Created Maven project with distributionManagement to Nexus, settings.xml with credentials, 3-stage Jenkinsfile. Pushed to GitHub, configured Jenkins SCM with PAT token, fixed Git tool path error 87, fixed Sonar 401 by using local token, fixed branch main vs master and git rebase issues. Final pipeline GREEN, artifact in Nexus, report in SonarQube."

■ Congratulations! You are now DevOps Engineer!